

Co-Frequency Interference Analysis Between Large-Scale NGSO Constellations and GSO Systems

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Abstract—With the number of multi-beam satellites in the non-geostationary orbit (NGSO) constellations increases by hundreds to thousands, the interference between geostationary orbit (GSO) satellite systems and NGSO constellations changes dynamically, and interference analysis will become very complicated. To address this problem, a dual-step interference analysis method based on space-discrete is proposed in this paper. Specifically, we discretizes the spatial positions of the reference NGSO satellite by dividing the latitude and longitude grids, and obtain the probability density function of the reference NGSO satellite. We analyze the aggregate interference caused by NGSO constellations to GSO systems, respectively, both in the uplink and downlink on the basis of interference-to-noise-ratio (I/N) and interference probability. The results show that there is severe interference between the NGSO and GSO systems and the aggregate interference level is affected by the locations of GSO earth station. Especially in the equatorial region, the interference is most serious. Furthermore, compared with commercial software Visualyse Professional, the proposed method can greatly improve the simulation efficiency while ensuring the accuracy.

Index Terms—Interference analysis, multi-beam satellite, large-scale non-geostationary orbit constellations.

I. INTRODUCTION

Non-geostationary orbit (NGSO) satellite systems, including low earth orbit (LEO) and medium earth orbit (MEO) systems, have been investigated since the 1990s. However, the unsuccessful commercial applications of the early NGSO satellite systems proposed decades ago have delayed further efforts to promote such systems for years. In recent years, with the development of micro satellite manufacturing technology and the reduction of satellite launch costs, a new wave of proposals for large-scale NGSO constellations of satellites using multi-beam have emerged [1]. For example, OneWeb is expected to launch 720 satellites with an orbital altitude of 1,200km [1]; more aggressively, Starlink has submitted satellite network filing of another 30,000 satellites [2] to the International Telecommunication Union (ITU), and is expected to launch a total of approximately 42,000 satellites with orbital altitudes varying from 340 to 1,325km.

There will be co-frequency interference between the GSO FSS and NGSO FSS systems when GSO and NGSO systems operating on the frequency band allocated to the fixed satellite service (FSS) at the same time. With the proposals of the large-scale NGSO constellations, the problem of NGSO constellations interfering with GSO satellite systems is becoming

increasingly prominent [3]. Moreover, the interference environment between the NGSO constellation and the GSO system is becoming complicated and the corresponding interference analysis is becoming challenging considering the huge number of satellites in the NGSO constellation. As a result, more research especially on large-scale NGSO satellite systems is required to evaluate the aggregate interference level between NGSO constellations and GSO satellite systems and provide guidance for the construction of satellite systems.

In recent years, there have been some breakthroughs in the study of interference analysis between GSO and NGSO systems. However most studies are based on time-discrete simulations and require lengthy time, which makes it no longer suitable for large-scale NGSO constellations [4]–[8]. Wang *et al.* analyzed the effect of exclusive angle on the equivalent power flux density (EPFD) for the downlink scenario where NGSO systems interfered with GSO satellite systems [4]; Li *et al.* analyzed the signal-to-interference-plus-noise-ratio (SINR) of the received signal by GSO earth station in the downlink interference scenario [5]; Zhang *et al.* analyzed NGSO system link availability by calculating carrier-to-noise-ratio (C/N) and interference-to-noise-ratio (I/N) for the downlink scenario where NGSO systems interfered with GSO satellite systems [6]; Sharma *et al.* studied both the uplink and downlink in-line interference between GSO and MEO satellite systems, and proposed a power control technique in order to mitigate the in-line interference [7]; Park *et al.* analyzed the effect of the number of NGSO satellites and the relative angle of NGSO satellites on the bit-error-rate (BER) performance [8].

To address the simulation time challenge, Fortes *et al.* proposed an analytical method to assess interference involving NGSO systems [9]. The rational behind it takes into account the fact that the positions of NGSO constellations can be determined once the position of one particular satellite (referred to as the reference satellite) is given. However, the simulation in [9] is relatively simple, which only adopts a single step size and does not apply frequency reuse scheme. Thus, more research especially on large-scale NGSO constellations based on multi-beam satellite is required to evaluate the aggregate interference. Motivated by the observations, a dual-step interference analysis method based on space-discrete is proposed in this paper to reduce the simulation running time under the condition of satisfying the required precision

of simulation. Different from the previous studies, which focus on relatively simple interference scenarios considering a few satellites with one beam in the NGSO constellation, we focus on the multi-beam satellites of the whole NGSO constellation while performing the aggregate interference analysis. The main contributions of our work are listed as follows:

- The uplink and downlink interference scenarios are introduced, respectively. Moreover, a mathematical model for calculating the aggregate interference is established given the fact that there will be multiple interference sources at the same time in the case of large-scale NGSO constellations.
- To reduce the time required for simulation while ensuring the simulation accuracy, a dual-step interference analysis method based on space-discrete is proposed. Moreover, the proposed method can perform a more accurate quantitative assessment of interference, since the whole constellation is considered.
- Taking OneWeb and SINOSAT-5 as the typical NGSO constellation and the typical GSO satellite system, respectively, the uplink and downlink interference analysis are performed and the impact of the earth station's geographic location on the interference level is analyzed by using both the proposed method and Visualyse Professional.

The remainder of this paper is organized as follows. In Section II, the system model with regard to multi-beam NGSO satellite is presented and the interference scenario is introduced. In Section III, the proposed interference analysis method is given. In Section IV, the simulation parameters are presented and the corresponding simulation results are discussed. Finally, we draw the conclusion and look forward to future work in Section V.

II. SYSTEM MODEL AND INTERFERENCE SCENARIO

In this paper, we consider a satellite system where the antenna of the earth station will always track the corresponding satellite to ensure the quality of the wanted signal [10]. The geometry used to evaluate the interference level is depicted in Fig. 1, which describes both the uplink and downlink interference between GSO and NGSO systems. In this paper, uplink interference means that the signal transmitted by the earth station belonging to the NGSO system is received by the satellite belonging to GSO system; downlink interference means that the signal transmitted by the satellite belonging to the NGSO system is received by the earth station belonging to GSO satellite system. In Fig. 1, T and R denote the transmitting earth station and the receiving earth station, respectively. It should be noted that in this paper, the transmitting earth station serves as gateway station and the receiving earth station serves as user terminal.

A. Single Entry Interference

The interference power is mainly determined by the transmission power, antenna gain and path loss. The uplink interference received at the satellite of the GSO satellite system is given by

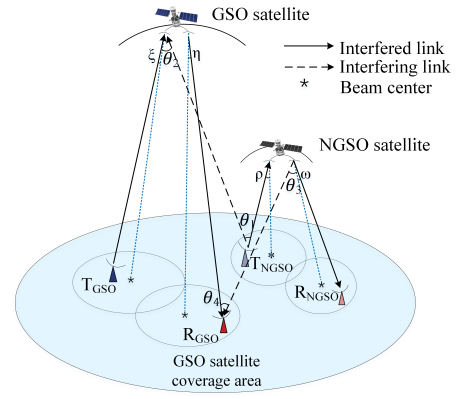


Fig. 1. The interference geometry between GSO and NGSO systems.

$$I_{up} = \frac{p_{t1} g_{ne} \theta_1 g_{gs} \theta_2 BAF_1}{\left(\frac{4\pi d_1}{\lambda_1} \right)^2}, \quad (1)$$

where p_{t1} denotes the gateway station transmission power of the NGSO system. $g_{ne} \theta_1$ and $g_{gs} \theta_2$ denote, respectively, antenna gains in directions θ_1 and θ_2 off the beam axis. BAF_1 indicates the bandwidth adjustment factor for uplink, $BAF_1 = \frac{Overlap_1}{B'_1}$, where B'_1 denotes the interfering system bandwidth in the uplink, $Overlap_1$ denotes the overlapping bandwidth of the interfering system and the interfered system in the uplink. d_1 denotes the distance corresponding to the uplink interference path and λ_1 indicates the wavelength corresponding to the uplink carrier frequency.

The downlink interference received by the user terminal of the GSO system is given by

$$I_{down} = \frac{p_{t2} g_{ns} \theta_3 g_{ge} \theta_4 BAF_2}{\left(\frac{4\pi d_2}{\lambda_2} \right)^2}, \quad (2)$$

where p_{t2} denotes the satellite transmission power of the NGSO system. $g_{ns} \theta_3$ and $g_{ge} \theta_4$ denote, respectively, the antenna gain in the directions θ_3 and θ_4 off the beam axis. BAF_2 is the bandwidth adjustment factor for downlink, the calculation is similar to BAF_1 . d_2 is the distance corresponding to downlink interference path and λ_2 indicates the wavelength corresponding to the downlink carrier frequency.

B. Aggregate Interference

Given the fact that emerging NGSO constellations have a large number of multi-beam satellites and apply frequency reuse [11], the aggregate interference should be considered when performing interference analysis.

Fig. 2 illustrates the interference scenario involving multiple interfering satellites with multi-beam, each with multiple interfering links. It is assumed that a four-color spectrum reuse pattern is applied in the interference scenario. However, for illustrative purposes, Fig. 2 only depicts beam with the same frequency (for example, the red beam) in the NGSO system.

I/N is an important metric in interference evaluation, which is used to model how the interference level vary in time and

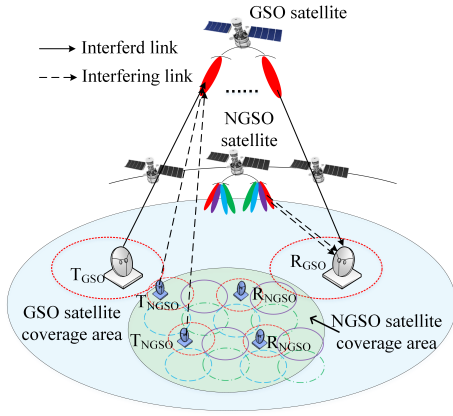


Fig. 2. The interference scenario involving multiple NGSO satellites.

space. The aggregate I/N of uplink and downlink are shown in (3) and (4), respectively:

$$\left(\frac{I}{N}\right)_{agg,up} = \frac{\sum_{s=1}^S \sum_{l=1}^{L_s} I_{ls,up}}{KB_1T_1}, \quad (3)$$

$$\left(\frac{I}{N}\right)_{agg,down} = \frac{\sum_{s=1}^S \sum_{l=1}^{L_s} I_{ls,down}}{KB_2T_2}, \quad (4)$$

where $I_{ls,up}$ and $I_{ls,down}$ represent the uplink and downlink single entry interference, respectively. S denotes the number of interfering satellites and L_s denotes the number of interfering links of satellite s . K represents the Boltzmann constant. B_1 and B_2 represent the uplink and downlink bandwidth of the interfered system, respectively. T_1 and T_2 represent the satellite receiver system noise temperature and the user terminal receiver system noise temperature in the GSO satellite system, respectively.

III. PROBABILITY BASED INTERFERENCE ANALYSIS METHOD

In this section, we provide a dual-step interference modelling method for the large-scale NGSO constellation, which not only evaluates the interference statistically but also optimizes the complexity of interference calculation. Specifically, a concise procedure for the aggregate interference calculation using the proposed method is given.

A. Probability Density Function of Reference NGSO Satellite

To obtain the PDF of the satellite position, we assume that the position of the reference satellite in a circular orbit is characterized by statistically independent random vector $x = \phi, \theta$. Given the fact that the reference satellite x may belong to two different orbital planes with the same inclination, we realize that the position of all other satellites in the constellation is a problem that has two possible solutions. In other words, there are two constellation configurations, namely ascending pattern and descending pattern. The criterion for judging that the constellation of the reference satellite is in

descending pattern is

$$\leq |v| < \frac{\pi}{2}, \quad (5)$$

where, v is true anomaly.

Since the position of the satellite in the constellation directly affects the interference power level, the ascending pattern and descending pattern should both be considered. Thus, the PDF of the satellite position is given by

$$p_x \phi, \theta \quad p_{x,AP} \phi, \theta \quad p_{x,DP} \phi, \theta, \quad (6)$$

where, $p_{x,AP} \phi, \theta$ and $p_{x,DP} \phi, \theta$ represent the probability density function of the satellite position in ascending and descending pattern, respectively. The $p_{x,AP} \phi, \theta$ and $p_{x,DP} \phi, \theta$ are equal, this is because the probability that the satellite belongs to the two orbits is the same.

According to [9], the PDF of the satellite position in a circular orbit is a function of the longitude ϕ and the latitude θ , which can be written as

$$p_x \phi, \theta = \begin{cases} \frac{1}{2\pi^2} \frac{\cos(\theta)}{\sqrt{\sin^2(\delta) - \sin^2(\theta)}}, & -\pi > \phi < \pi, -\delta > \theta < \delta \\ e \text{ se.} & \text{else.} \end{cases} \quad (7)$$

where, δ is the orbital inclination of the NGSO system.

B. Procedures to Determine the Statistical Interference

The reference satellite position of the NGSO constellation is scattered on the latitude and longitude plane according to a certain latitude and longitude grid resolution to form a set of latitude and longitude grids. For each element in the grid, it is assumed that the reference satellite is at its center. Then, the probability of the NGSO satellite appearing in the grid can be obtained by using (6) and (7).

Once the position of the reference NGSO satellite is known, the positions of other satellites in the NGSO constellation can be determined, and then the corresponding interference scenario will be determined. In other words, I/N is a deterministic function relative to the position of the reference satellite. Therefore, assuming that the other system parameters, such as the antenna pattern for the satellite and earth station, transmission power, antenna tracking strategy of the earth station, et., are known, we can derive the PDF and the cumulative distribution function (CDF) of I/N.

Considering the accuracy and complexity of the simulation, compared to coarse resolution, fine resolution is more suitable for detecting fast variations of the interference level, but at the same time it will cause longer simulation time. To address this problem, taking into account the accuracy and complexity of the simulation, we perform the interference analysis in two stages based on the fine-step range defined in [12]. In the first stage, a fine precision of latitude and longitude grid resolution is adopted in the fine-step range where the interference power have large variation. In the second stage, a coarse resolution is adopted outside the fine-step range because the interference level within this range changes gently.

TABLE I
SIMULATION PARAMETERS OF SINOSAT-5

Parameters	Values
Parameters for uplink	
Frequency band	28.05 GHz
Bandwidth	100 MHz
Satellite peak antenna gain	45 dBi
Satellite antenna half power beamwidth	0.9°
Satellite antenna pattern	Rec ITU-R S.672-4 [15]
System noise temperature	640 K
Parameters for downlink	
Frequency band	11.545 GHz
Bandwidth	60 MHz
User terminal peak antenna gain	62.4 dBi
User terminal antenna half power beamwidth	0.13°
User terminal antenna pattern	AP8 [16]
System noise temperature	240 K
Protection criteria	
I/N	-12.2 dB [17]

TABLE II
SIMULATION PARAMETERS OF ONEWEB

Parameters	Values
Parameters for uplink	
Frequency band	28.05 GHz
Bandwidth	100 MHz
Transmit power of gateway	29 dBW
Gateway peak antenna gain	45 dBi
Gateway antenna half power beamwidth	0.9°
Gateway antenna pattern	AP8 [16]
Parameters for downlink	
Frequency band	11.5125 GHz
Bandwidth	125 MHz
Transmit power of satellite	7.5 dBW
Satellite peak antenna gain	25.9 dBi
Satellite antenna half power beamwidth	7.9°
Satellite antenna pattern	Rec ITU-R S.1528 [18]

IV. SIMULATION AND ANALYSIS

In this section, using OneWeb interference SINOSAT-5 as a simulation case, the proposed method and Visualyse Professional are both employed to analyze the interference of the NGSO constellation to the GSO satellite system.

A. GSO and NGSO System Parameters

The longitude of the SINOSAT-5 satellite is 110.5°E. By searching the satellite network list (SNL) released by ITU, the SINOSAT-5 SNL ID is 106520145 [13], in which the uplink beam RA1R and downlink beam EKU1 are selected to determine the related communication parameters. The SINOSAT-5 system parameters are listed in Table I.

OneWeb's Ku-Ka-band constellation comprises 720 satellites distributed in 18 circular orbital planes at an orbital altitude of 1,200 km, 87.9° inclination. By searching the SNL released by ITU, the OneWeb SNL ID is 113520120 [14], in which the uplink beam GRA and downlink beam TAR are

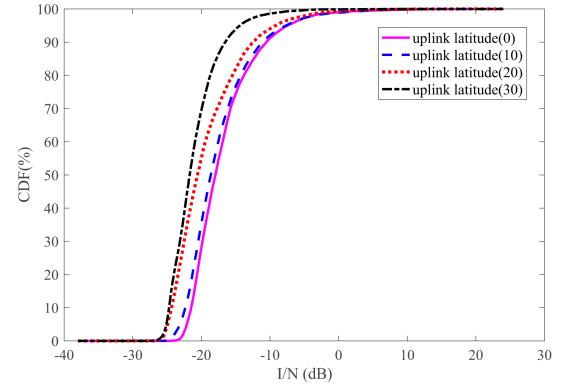


Fig. 3. The CDF of I/N for uplink by Visualyse Professional.

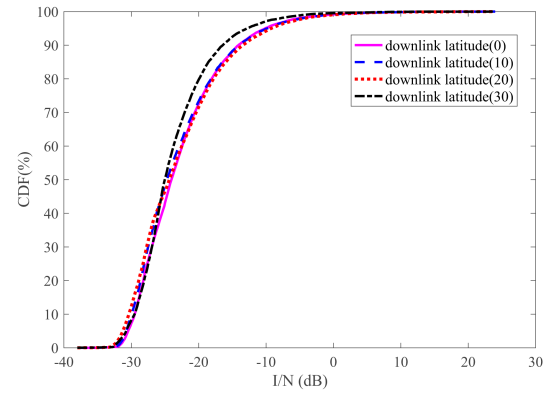


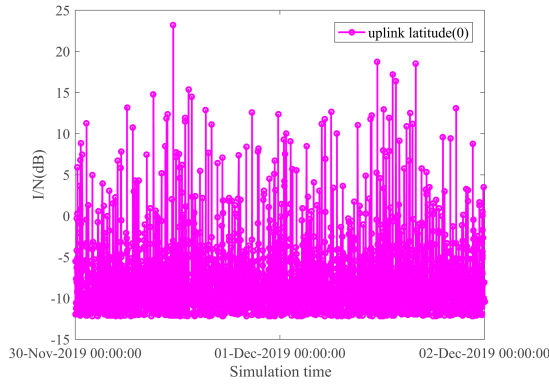
Fig. 4. The CDF of I/N for downlink by Visualyse Professional.

selected to determine the related communication parameters. The OneWeb system parameters are listed in Table II.

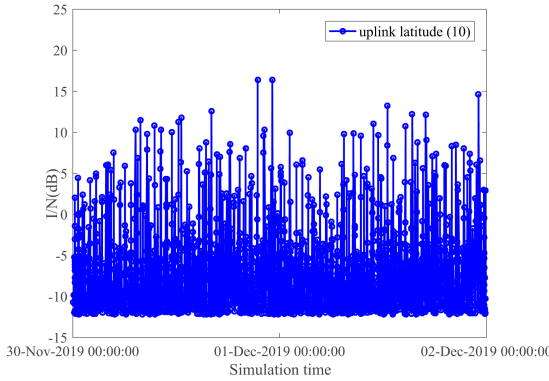
B. Simulation Results by Visualyse Professional

When using Visualyse Professional, the simulation time step is 10 seconds and the total simulation time is 3 days. In the uplink, a total of 41 OneWeb gateways are deployed with an inter-station distance of 800km, using GSO gateway station as the center. In the downlink, a total of 122 OneWeb user terminals are deployed with an inter-station distance of 50km, using GSO user terminal as the center.

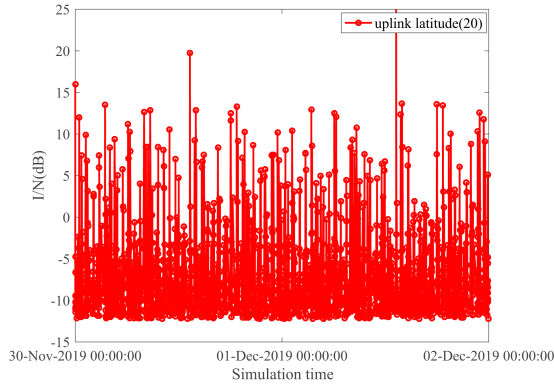
In Fig. 3 and Fig. 4, we demonstrate the interference levels in the uplink and the downlink with different GSO earth station latitudes, respectively, where the longitude of the GSO earth station is 110.5°E. In this paper, the uplink is the feeder-link and the downlink is the user-link. It can be seen from Fig. 3 that, the I/N will exceed the interference threshold. Moreover, in the range of 0 to 30°N, the higher latitude of the GSO gateway station, the lower level of the interference. This is because, in the range of 0 to 30°N, with the increase of GSO gateway latitude, the change of the off-axis angle at the NGSO gateway is small with almost the same transmit gain; however, the off-axis angle at the GSO satellite increases with the decreased receive gain. In comparison, in Fig. 4, lower satellite transmission power can cause lower interference from the NGSO satellite to the NGSO user terminal.



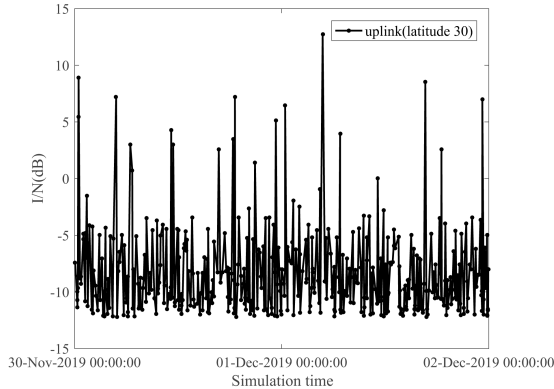
(a) The latitude of the GSO gateway station is 0° .



(b) The latitude of the GSO gateway station is 10° .



(c) The latitude of the GSO gateway station is 20° .



(d) The latitude of the GSO gateway station is 30° .

Fig. 5. The interference level and time distribution.

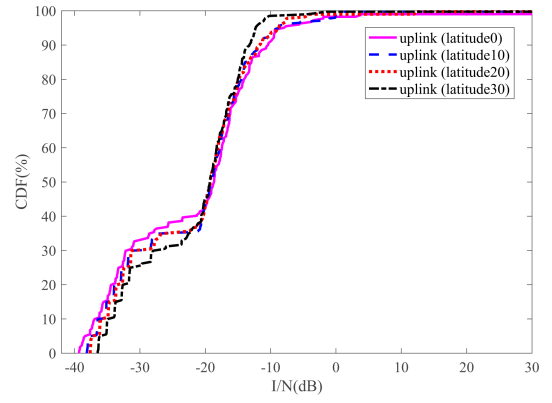


Fig. 6. The CDF of I/N for uplink obtained with the proposed method.

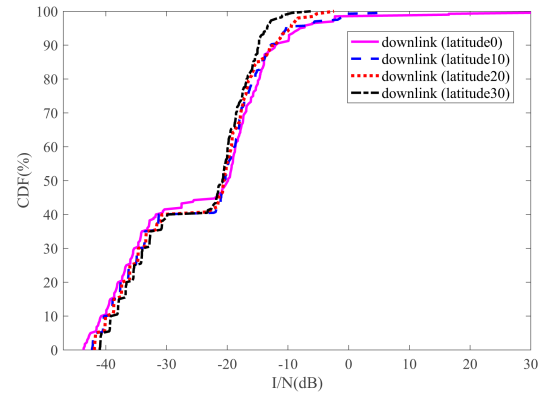


Fig. 7. The CDF of I/N for downlink obtained with the proposed method.

In Fig. 5, we investigate the time distribution of the interference events with I/N exceeding -12.2 dB in the uplink. As shown in Fig. 5, in the range of 0 to 30° N, the higher latitude of the GSO gateway, the smaller number of interference events, which is consistent with our findings in Fig. 3. Considering that the trend of the downlink is similar to the uplink, the interference time distribution of the downlink is not given here due to space limitation.

C. Simulation Results by Proposed Method

In Fig. 6 and Fig. 7, we investigate the interference levels in the uplink and the downlink with different GSO earth station latitudes, respectively. The interference levels in Fig. 6 and Fig. 7 obtained by the proposed method are similar to the simulation results obtained by Visualyse Professional.

For a clearer comparison, we focus on the worst uplink situation, in which the latitude of the GSO gateway station is 0° . As shown in Fig. 8, in the range of $I/N > -20$ dB, a almost agreement between the results produced by Visualyse Professional and the proposed method can be obtained and the probability of $I/N > -12.2$ dB is 14.55%. Moreover, we note that the simulation results of the proposed method range from -39 to 20 dB, and the simulation results of Visualyse Professional mainly fall in the range of -22 to 20 dB. The difference between the results is due to the fact that the 3-

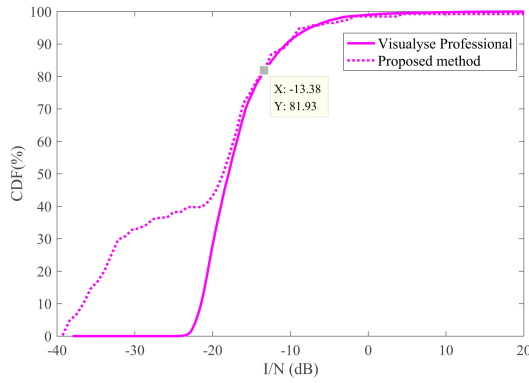


Fig. 8. The CDF of I/N for uplink obtained by the proposed method and Visualyse Professional. (The latitude of the GSO gateway station is 0°)

day simulated time using Visualyse Professional is too short compared with the proposed method which correspond to the infinite number of simulated time.

In the case of uplink with a GSO gateway station latitude being 0° , it takes around 38 minutes using Visualyse Professional, and approximately 25 seconds to generate the results using the proposed method. It can be seen that the proposed method can improve the simulation efficiency by about 80 times, which can be explained from the following reasons. On the one hand, as a analytical method, the proposed method is decoupled from the simulation time and the time required for simulation is mainly related to the number of latitude and longitude grids. On the other hand, due to a dual-step is used in simulation, the number of divided latitude and longitude grids can be further reduced. Therefore, the proposed method will be more suitable for the complicated scenarios involving a large number of earth stations and NGSO satellites.

V. CONCLUSION

In this paper, we have analyzed the co-frequency interference between large-scale NGSO and GSO systems both in the downlink and uplink, respectively. First, a mathematical model for aggregate interference analysis is established. Then, to improve the simulation efficiency, a dual-step interference analysis method based on space-discrete is proposed. Based on the results obtained from the simulation, we observe that the I/N will exceed the interference threshold. Several simulations considering different latitudes of GSO earth stations being $0^\circ/10^\circ/20^\circ/30^\circ$ are performed in this paper. The results show that when the GSO earth station moves from 30°N to the equator, the interference received by the GSO system becomes worse and in the worst uplink situation the probability of $I/N > -12.2\text{ dB}$ is 14.55%. Finally, considering that NGSO constellations should not cause harmful interference to GSO satellite systems in the scenario of frequency sharing between GSO and NGSO systems, interference mitigation techniques to promote harmonious spectrum coexistence should be studied in our future work.

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